

Functional Requirement Document

Here is the detailed Functional Requirement Document (FRD) based on the provided Business Requirement Document (BRD) and adhering to the "Coding Requirements" section.

FR-DLT-001: Load Customer and Order Data into Delta Tables

Title: Load Customer and Order Data into Delta Tables

Description: The system must read source CSV data from a specified volume, load it into delta tables for customer and order data, and perform necessary data transformations and validation.

Preconditions:

- Source CSV files for customer and order data exist in the specified volume.
- The delta tables for customer and order data do not have any existing data that needs to be preserved.

Main Flow / Functional Steps:

- Read customer CSV data from ``/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata``.

Load customer data into a delta table with schema: ``CustId`, `Name`, `EmailId`, `Region``.

- Read order CSV data from ``/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata``.

Load order data into a delta table with schema: ``OrderId`, `ItemName`, `PricePerUnit`, `Qty`, `Date`, `CustId``.

- Add a new column ``TotalAmount`` to the order delta table, calculated as the product of ``PricePerUnit`` and ``Qty``.

FR-DLT-002: Clean and Transform Customer and Order Data

Title: Clean and Transform Customer and Order Data

Description: The system must remove null and duplicate records from the customer and order delta tables.

Preconditions:

- Customer and order delta tables exist and have data loaded.

Main Flow / Functional Steps:

- Remove null records from the customer delta table.
- Remove duplicate records from the customer delta table.

- Remove null records from the order delta table.
- Remove duplicate records from the order delta table.

FR-DLT-003: Create and Load ordersummary Table

Title: Create and Load ordersummary Table

Description: The system must create an `ordersummary` table if it does not exist, and load it with joined customer and order data using SCD type 2.

Preconditions:

- Customer and order delta tables exist and have clean data.
- `ordersummary` table does not exist or is empty.

Main Flow / Functional Steps:

Create `ordersummary` table with schema: `CustId`, `Name`, `EmailId`, `Region`, `OrderId`, `ItemName`, `PricePerUnit`, `Qty`, `Date` if it does not exist in `catalog="gen\_ai\_poc\_databrickscoe"` and `schema="sdlc\_wizard"`.

- Join customer and order delta tables on `CustId`.
- Load joined data into `ordersummary` table using SCD type 2.

FR-DLT-004: Update ordersummary Table on Customer Changes

Title: Update ordersummary Table on Customer Changes

Description: The system must update the `ordersummary` table whenever there is a change in the customer table, maintaining history using SCD type 2.

Preconditions:

- `ordersummary` table exists and has data.
- Customer delta table has changes.

Main Flow / Functional Steps:

- Identify changes in the customer delta table.
- Update `ordersummary` table by making old records inactive and new records active.
- Update `StartDate` and `EndDate` accordingly to maintain history.

FR-DLT-005: Create and Load customeragggregatespend Table

Title: Create and Load customeragggregatespend Table

Description: The system must create a `customeraggregatespend` table if it does not exist, and load it with aggregated data from `ordersummary`.

Preconditions:

- `ordersummary` table exists and has data.
- `customeraggregatespend` table does not exist or is empty.

Main Flow / Functional Steps:

Create `customeraggregatespend` table with columns: `Name`, `TotalAmount`, and `Date` if it does not exist in `catalog="gen\_ai\_poc\_databrickscoe"` and `schema="sdlc\_wizard"`.

- Aggregate `TotalAmount` from `ordersummary` table, grouping by `Name` and `Date`.
- Load aggregated data into `customeraggregatespend` table.

FR-DLT-006: Implement Delta Live Tables

Title: Implement Delta Live Tables

Description: The system must implement the above requirements using Delta Live Tables.

Preconditions:

- All previous requirements have been implemented.

Main Flow / Functional Steps:

- Implement steps 1 to 5 using Delta Live Tables.